# Chapter 4 Simple Regression

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Statistical Decision Making (45-752)

[1]:
```
# Jupyter "magic" command
# precision - round floats to 4 significant digits
%precision %.4f

import tprstats
import pandas
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import t
```

## 1 Simple Regression

### 1.0.1 Learning objectives:

After completing this chapter, you will be able to:

- Explain how a simple regression model represents a relationship between variables.
- Estimate a simple regression model using data.
- Interpret the results of a simple regression model.
- Make predictions using a simple regression model.

### 1.1 Simple Regression: Population Model

The simple regression model is a relationship between a variable $y$ and a variable $x$. We write the population model as follows:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

We refer to $x$ as the **independent variable**, also sometimes called the **explanatory variable**. We refer to $y$ as the **dependent variable**. We denote the unobserved random component $\varepsilon$. The unknown coefficients $\beta_0$ and $\beta_1$ are the **population coefficients**. We denote the population coefficients with Greek betas as in Equation (1) above. In statistics, an unknown constant is called a **parameter**. Our equation above has two parameters, $\beta_0$ and $\beta_1$. The above model is called the simple regression model because there is only one explanatory variable. In the simple regression

model, an **observation** is a pair of numbers $(x_i, y_i)$ generated by the model. We estimate the model using our sample of data which will contain multiple observations. The number of observations in a sample is denoted $n$. The index $i$ takes on values $i = 1, 2, ..., n$.

When we conduct statistical analysis, we typically develop a population model that we believe describes the problem we wish to study. We don't know the population parameters, so we must estimate our model using a **sample** of data. In some situations, we undertake an experiment to obtain data. In managerial applications, we often use data that have been generated in the managerial environment we are studying. For example, later in the course, we will study a statistical model developed by a Tepper alum to help a major car manufacturer quantify how the prices of tires that it purchases vary with tire characteristics.

Using the sample of data, we estimate the unknown population parameters. In the first application below, we study the relationship between acres of forest burned in the US and global surface temperature of the earth. Our population model for this application is given below. The dependent variable, millions of acres of US forest burned, is denoted $Acres$. The independent variable, global average surface temperature in degrees Fahrenheit, is denoted $Tempf$.

$$Acres_i = \beta_0 + \beta_1 Tempf_i + \varepsilon_i$$

## 1.2 Simple Regression: Implementation

### 1.2.1 Obtaining Data

Obtaining and organizing data is rarely easy. Often extensive review of an organization's records is required, and it is frequently necessary to consolidate data from multiple sources. You will see from the footnotes in this chapter the sources I found in searching for the data for the first three applications. The fourth application entailed assembling data from more than one administrative department in Tepper.

We will use data from 1985 through 2020. Hence, our number of observations is 56, i.e., $n = 56$. To help you understand the terminology and notation, the first five observations are show below. For the first observation, $x_1 = 57.31$ and $y_1 = 2.896$. For the fourth observation, $x_4 = 57.74$ and $y_4 = 5.009$.

| $i$ | Acres | Tempf |
|---|---|---|
| 1 | 2.896 | 57.31 |
| 2 | 2.719 | 57.45 |
| 3 | 2.447 | 57.72 |
| 4 | 5.009 | 57.74 |
| 5 | 1.827 | 57.56 |

We often wish to refer to the set of values of a variable. For example, in our model above, we use $Acres$ to denote the vector of 56 values of the dependent variable, $Tempf$ to denote the vector of 56 values of the independent variable, and $\varepsilon$ to denote the vector of 56 error terms in the regression.

For your review:

- On which side of the regression equation is the independent variable?

- What is the dependent variable in the equation relating Acres and Tempf?
- What is the difference between an observation and a sample?
- Is $\varepsilon$ a parameter? Hint: Is $\varepsilon$ an unknown constant?
- Does "explanatory variable" mean the same thing as "independent variable"? Hint: Yes:)

### 1.2.2 Estimation

In regression, the estimated coefficients are chosen to minimize the sum of squared estimated errors. When we estimate a simple regression, we obtain the estimated intercept, $b_0$, and the estimated slope, $b_1$.[1] Using these estimated coefficients, we calculate the predicted value for each observation:

$$\hat{y}_i = b_0 + b_1 x_i$$

We can then calculate the estimated error term, $e_i$, for each observation:

$$e_i = y_i - \hat{y}_i$$

In the applications below, we will estimate the coefficients using Python, and we will calculate the predicted values $\hat{y}_i$ and the estimated error terms $e_i$. The estimated error terms are called **residuals**.

### 1.2.3 Notation

In the population model introduced earlier in this chapter, we used Greek letters $\beta_0$ and $\beta_1$ to denote the unknown population coefficients. To distinguish between the population coefficients and the estimated coefficients, we denote the latter as $b_0$ and $b_1$. Please remember this notational convention: $\beta_j$ denotes an unknown *population* coefficient and $b_j$ denotes an *estimated* coefficient. Note also that $\varepsilon_i$ denotes the error term for observation $i$ in the population model and $e_i$ denotes the estimated error term for observation $i$.

## 1.3 Simple Regression Properties

In the probability and statistics course, the ideal conditions for regression were introduced, and you learned the Gauss-Markov Theorem. Here we state the assumptions underlying the Theorem and then interpret properties established by the theorem.

Suppose the following assumptions are satisfied:

1. The $x_i$ are fixed for all observations $i$. This means the $x_i$ are not randomly determined. For example, calendar time is not a random variable. Later we will consider applications where the $x_i$ may be random.
2. The $\varepsilon_i$ are from a population with a mean of zero.
3. The $\varepsilon_i$ are from a population with a constant variance $\sigma^2$: $E(\varepsilon_i) = 0$, $Var(\varepsilon_i) = \sigma^2$.
4. The error terms are not correlated with each other: $Cov(\varepsilon_i, \varepsilon_j) = 0$ for all $i \neq j$.

[1] An optional appendix to this chapter provides a mathematical derivation of the formulas used to estimate the coefficients.

### 1.3.1 Gauss Markov Theorem

The Gauss Markov Theorem establishes that, under the above assumptions, least squares is the Best Linear Unbiased Estimator of the coefficients. This often goes by the shorthand that least squares is BLUE. In Chapter 6, we introduce the multiple regression model. In that Chapter, a detailed interpretation of the above conditions is provided.

## 1.4 Plan For Analyses That Follow

### 1.4.1 Topics We Will Investigate

You have surely seen news reports of the extensive destruction of US forests caused by forest fires. Below, we study this problem via three applications of the simple regression model. Our fourth application uses data for the Tepper School.

1. We study the relationship between acres burned and time.
2. We study the relationship between world average surface temperature and time.
3. We study the relationship between acres burned and world surface temperature.
4. We study the relationship of GMAT and graduating GPA in required courses at Tepper.

### 1.4.2 Overview of Data

For our first three applications, I assembled a data set that contains annual data for the following five variables: Year, Acres, Fires, Tempc, and Tempf. The variables are defined as follows:

- Year is calendar year.
- Acres is the number of millions of acres of US forest burned each year.
- Fires is the number of US forest fires during the year.
- Tempc is global surface temperature in degrees Celsius.
- Tempf is global surface temperature in degrees Fahrenheit.

We will use simple regression models to study these data. The data are in a file named: Fire_and_Temperature_Data.xlsx. We import the data into JupyterLab using the command below.

[2]:
```
Fire_and_Temperature_Data = pandas.read_excel("data/Fire_and_Temperature_Data.
↪xlsx")
```

### 1.4.3 Global Surface Temperature Since 1880

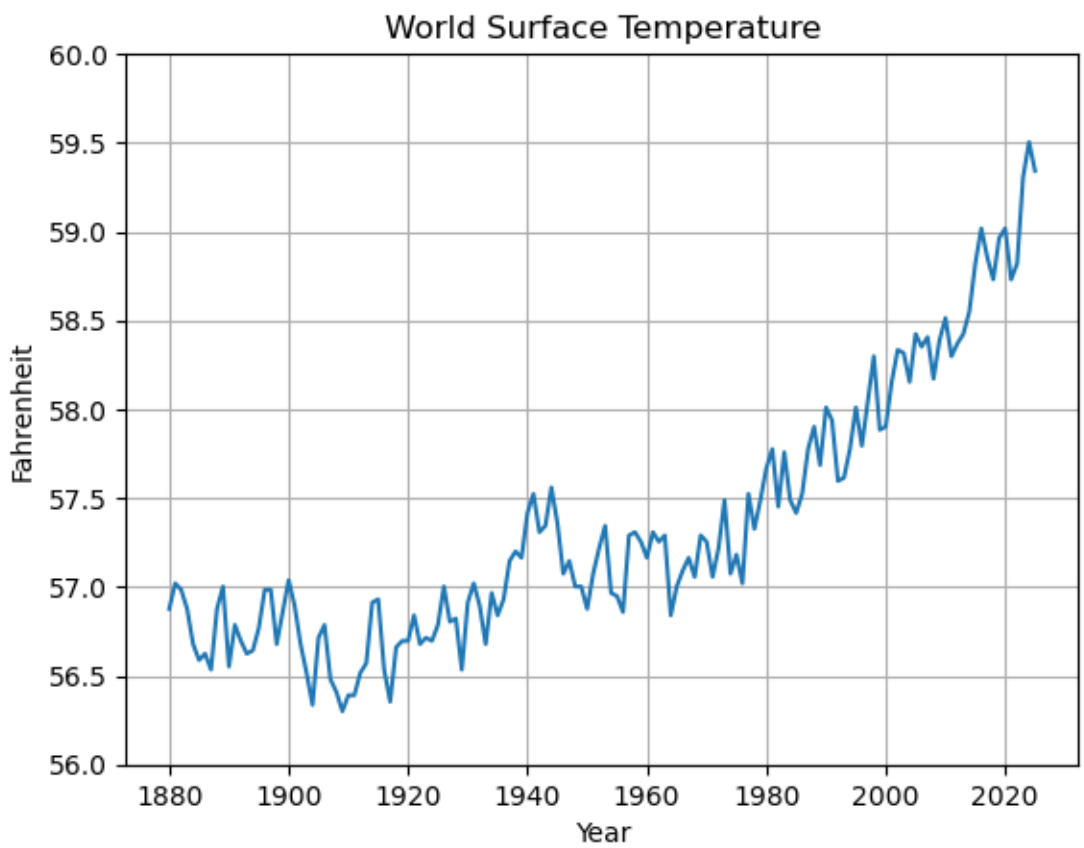
The following graph provides a long-term perspective on Temperature[2]. With a bit of trial-and-error, I found that setting the values 56 to 60 provided a good range for the vertical axis. In the command below, I set this range using plt.ylim(56,60). The time period for our data is 1880 to 2025. Temperatures are in degrees Fahrenheit.

Viewing this graph, we see that there was a slight downward trend from 1880 to 1910. This was followed by an upward trend that is ongoing. The peak in temperatures during the early 1940's is due to measurement error. WW2 did not cause an increase in global temperatures but it did result in substantial measurement error in temperatures.

[2] https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/global/time-series

[3]:
```
model = tprstats.model("ts", "Tempf ~ Year", Fire_and_Temperature_Data)

plt.plot(model.X.iloc[:, 1:], model.y)
plt.title("World Surface Temperature")
plt.xlabel("Year")
plt.ylabel("Fahrenheit")
plt.ylim(56, 60)
plt.grid(True)
```




Data for forest area burned in the US are available from 1926 onward. However, the accuracy of measurement of area burned was quite problematic. Systematic retrospective measurment has been undertaken in 1984. Hence, we will use data from 1984 onward.

[4]:
```
FTD84 = Fire_and_Temperature_Data.query("Year>1983")
```

### 1.4.4 Looking At The Data

As you know, it is always a good idea to begin by looking at the data. By doing so, we get a feel for our data, and we may detect any potential anomalies or coding errors in the data. We will look at the data in several ways.

**Summary Statistics** Let's begin with summary statistics. We see from the summary statistics below that the area of forest burned between 1985 and 2020 has ranged between 1.3 million acres and 10.1 million acres. Scroll down and you will see that the acreage burned was just over 10 million in 2015 and again in 2017. Thus, in these two years combined, the forest acreage burned in the United States was equal to roughly two thirds of the area of Pennsylvania!

We also see that temperature varied between 57.31 and 58.87 degrees Fahrenheit (14.06 to 14.93 degrees Celsius) over the 1985-2020 time period. Looking at the data, we see that the temperature in 1985 was 57.31, the lowest temperature over this 36 year period and the highest temperature over this 36 year period was 58.87 in 2020. Is it just coincidental that the lowest temperature was at the beginning and the highest temperature at the end, or is there a time pattern in the data? We will investigate further below.

[5]:
```
FTD84.describe()
```

[5]:

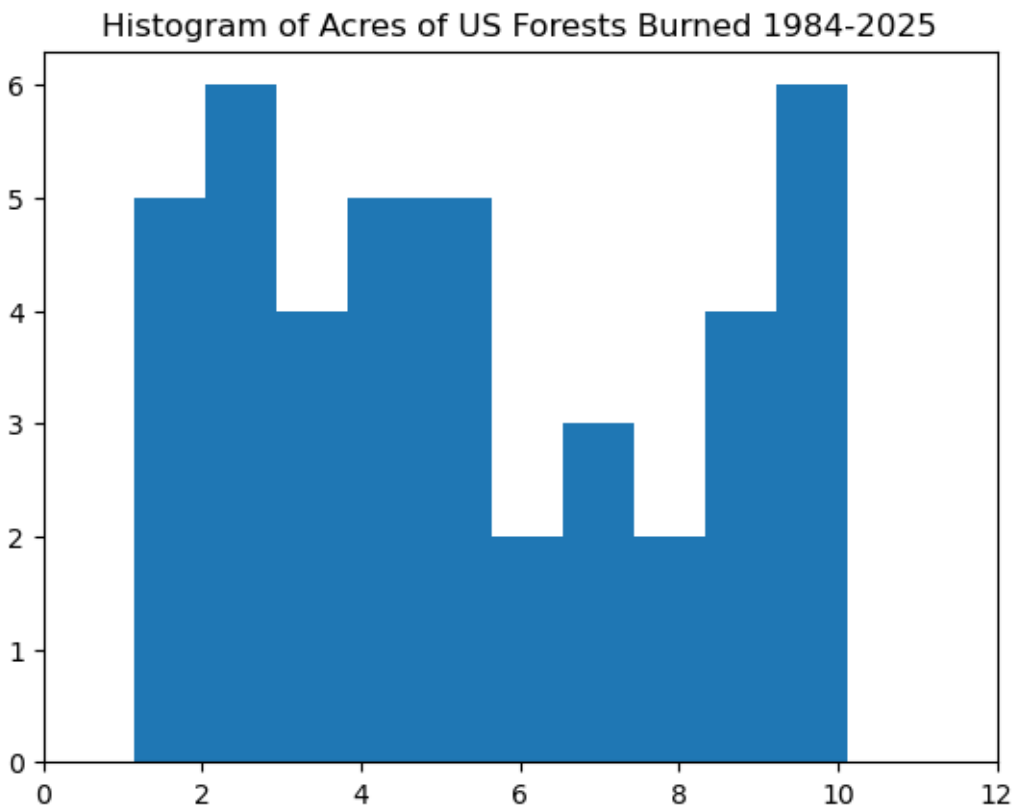
| | Year | US_Acres_Burned | Fires | Tempc | Tempf |
|---|---|---|---|---|---|
| count | 42.000000 | 42.000000 | 42.000000 | 42.000000 | 42.000000 |
| mean | 2004.500000 | 5.443821 | 70959.857143 | 14.611429 | 58.300571 |
| std | 12.267844 | 2.829623 | 14663.964260 | 0.294358 | 0.529845 |
| min | 1984.000000 | 1.148409 | 20493.000000 | 14.120000 | 57.416000 |
| 25% | 1994.250000 | 2.910505 | 63946.000000 | 14.390000 | 57.902000 |
| 50% | 2004.500000 | 5.070382 | 71399.500000 | 14.615000 | 58.307000 |
| 75% | 2014.750000 | 7.967706 | 80559.000000 | 14.825000 | 58.685000 |
| max | 2025.000000 | 10.125149 | 96385.000000 | 15.280000 | 59.504000 |

**Histograms** Let's view a histogram of acres burned to get a feel for the distribution of acreage burned. Note that I used the xlim option to set the range of the bottom axis. I chose this range after a bit of trial and error. Study the format of the command because it is often useful to set the range of the horizontal axis on your graph. Issue the hist() command without specifying xlim to see the difference.

[6]:
```
plt.hist(FTD84["US_Acres_Burned"])
plt.title("Histogram of Acres of US Forests Burned 1984-2025")
```

```
plt.xlim(0, 12)
```

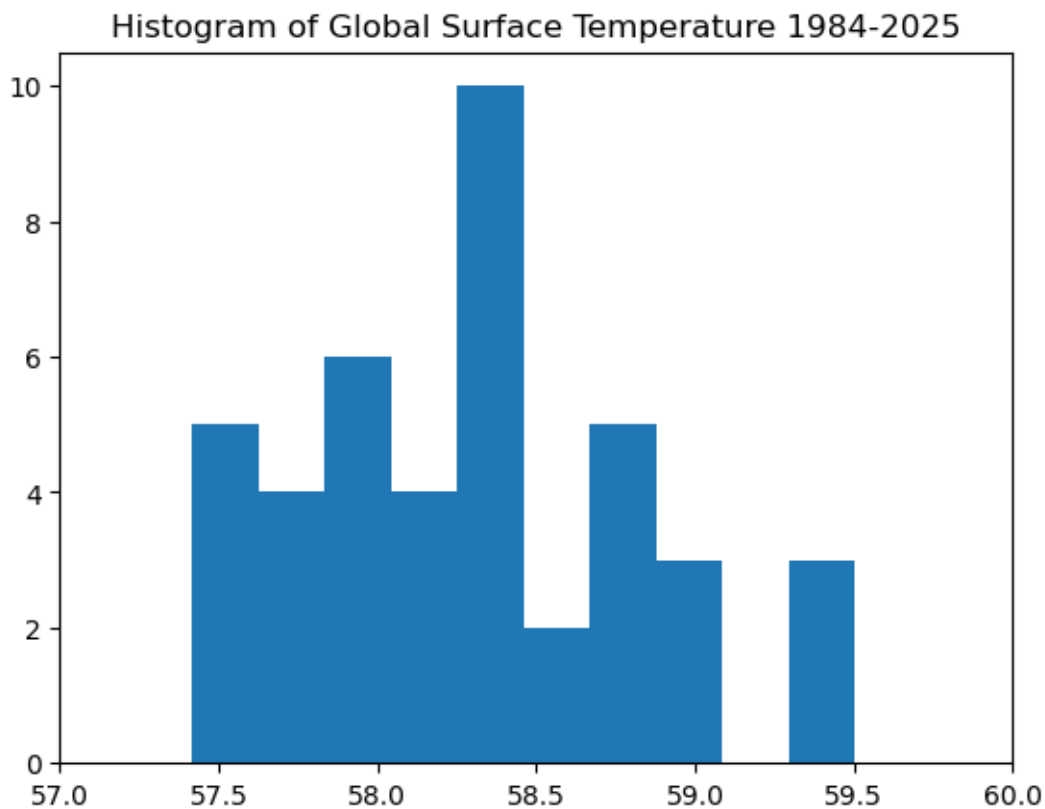
[6]: (0.0000, 12.0000)




Next, we view a histogram of temperatures. We saw from the summary statistics that temperatures ranged from roughly 57.31 to 58.87 over the time period in our data. The histogram below summarizes the distribution of temperatures over our 36 year time period.

[7]:
```
plt.hist(FTD84["Tempf"])
plt.title("Histogram of Global Surface Temperature 1984-2025")
plt.xlim(57, 60)
```

[7]: (57.0000, 60.0000)




## 1.5 Four Applications

We now study four applications of the simply regression model. These applications will help you learn how to interpret and evaluate regression models. In addition, I think you will find the results of these analyses to be very interesting.[3]

### 1.5.1 Application 1: Forest Area Burned As A Function of Time

When beginning an analysis of the relationship between two variables, it is useful to think about the algebraic sign that you expect. Do you expect a positive or negative relationship, or no relationship. That way, you are more likely to recognize a situation in which the outcome is not what you expected. What algebraic sign would you expect for the slope coefficient of the relationship of Acres and Year? That is, do you expect Acres burned to be increasing over time, decreasing, or no relationship?

**Estimate Regression of Acres as a Function of Year** The regression is estimated by the first line below. We view the regression results with the second command. Since this a time series, we use "ts" rather than "cs" in the command. Also, you will see that the first note following the

[3]The inspiration for the applications is the work of Marshall Burke, Sam Heft-Neal, and Michael Wara, "Managing The Growing Cost of Wildfires", Stanford Institute for Economic Policy Research, October 2020. I thank them for sharing their data with us.

regression provides information about how the standard errors are calculated. Don't worry about these now. We will cover these in detail later in the course. The second note following the regression is meaningless in this regression, so just ignore it.

[8]:
```
RegAcres = tprstats.model("ts", "US_Acres_Burned~Year", FTD84)
RegAcres.summary()
```

[8]:

| | | | |
|---|---|---|---|
| **Dep. Variable:** | US_Acres_Burned | **R-squared:** | 0.346 |
| **Model:** | OLS | **Adj. R-squared:** | 0.329 |
| **No. Observations:** | 42 | **F-statistic:** | 30.81 |
| **Covariance Type:** | HAC | **Prob (F-statistic):** | 2.02e-06 |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **Intercept** | -266.4386 | 48.853 | -5.454 | 0.000 | -365.174 | -167.703 |
| **Year** | 0.1356 | 0.024 | 5.551 | 0.000 | 0.086 | 0.185 |

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 2 lags and without small sample correction
[2] The condition number is large, 3.32e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

**Interpreting The Four Columns of The Regression Output** We will now interpret the regression output in detail. Looking back at the regression, we see four columns of numbers labeled as follows: coef, std err, t, P>|t|.

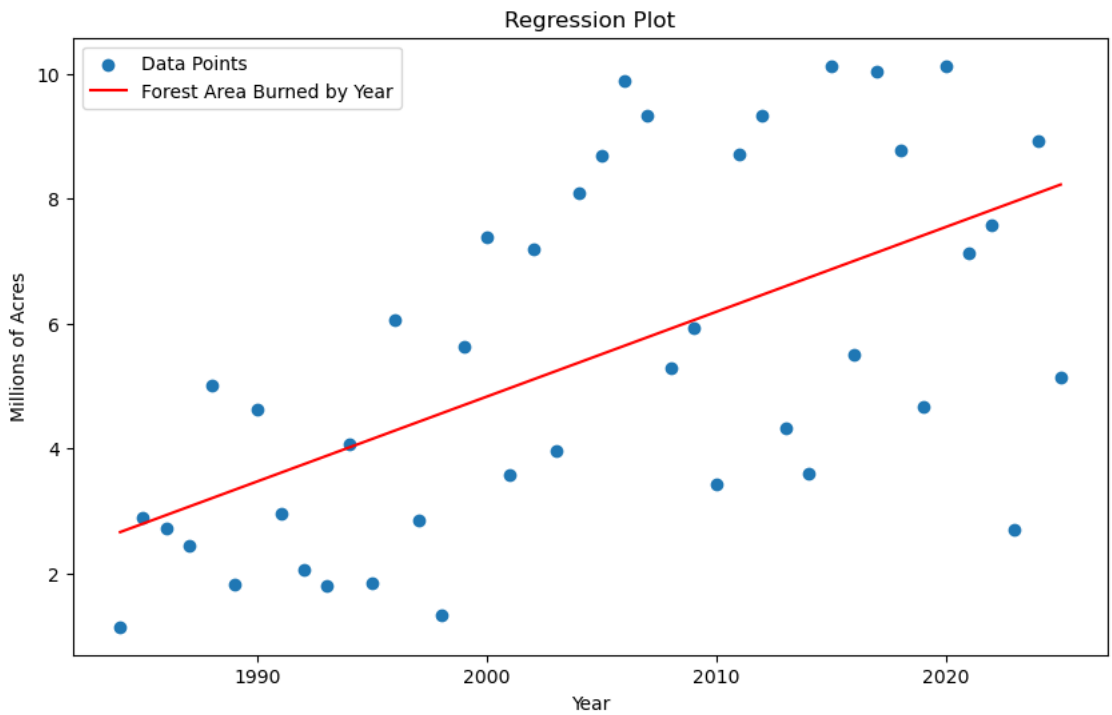
- The coef column provides the estimated coefficients. Recall that Acres is measured in millions. Thus, the slope coefficient of the regression tells us that estimated area burned increased by 135,600 acres per year. You will probably be uneasy about the negative intercept. Don't worry about that for the moment. We will discuss it a bit later below.
- The std err column provides the estimated standard error for each of the estimated coefficients. The standard error of a coefficient is an estimate of the accuracy of the estimate of the coefficient.
- The t value column in the regression output is the ratio of each coefficient to its standard error, i.e., the ratio of the first column to the second column. The logic is explained below.
- The column P>|t| provides the p-value for each coefficient under the null hypothesis that the corresponding population coefficient is zero. The very small p-values for both coefficients in the above regression tell us that we strongly reject the null hypothesis that the associated population coefficients are zero.

**Plotting The Data And The Regression Line** Plots are often a useful way to look at data. We will do plots for each of the four applications that we study. Let's look at a plot of acres burned versus year with the regression line shown on the plot.

[9]:
```
# Predicted values
y_pred = RegAcres.predict()

# Plotting
plt.figure(figsize=(10, 6))
```

```
plt.scatter(FTD84["Year"], RegAcres.y, label="Data Points")
plt.plot(FTD84["Year"], y_pred, color="red", label="Forest Area Burned by Year")
plt.xlabel("Year")
plt.ylabel("Millions of Acres")
plt.title("Regression Plot")
plt.legend()
plt.show()
```




**Testing A Hypothesis about a Coefficient in a Regression Model** The coefficient for Year tells us that area burned is estimated to increase by 164,400 acres per year. Let's work through how we test a hypothesis about a population regression coefficient. The t statistic to test a null hypothesis about the slope coefficient is:

$$t = \frac{b_1 - \beta_{1,0}}{s_{b_1}}$$

In the above expression, $\beta_{1,0}$ denotes the null hypothesis for the slope coefficient. It is a number that you specify when you test a null hypothesis. The term in the denominator, $s_{b_1}$ is estimated standard error of the estimated slope coefficient. The standard error is reported under the column "Std. Error" in the regression output.

As an example, suppose we wish to test the null hypothesis that the population slope coefficient in the above regression model is $\beta_{1,0} = .2$, which would imply that acres burned is increasing by 200,000 per year. Let's use significance level $\alpha = .05$. Our calculated t-statistic is:

$$t = \frac{.1356 - .2}{.024} = -1.548$$

In a regression, the number of **degrees of freedom** is the number of observations minus the number of coefficients estimated. We denote this with Greek letter $\nu$. We find the degrees of freedom as follows. Our data are for 1984 through 2025, a total of $n = 42$ observations. We estimated two coefficients, the intercept and slope. Hence, our degrees of freedom, $\nu$ equals $42 - 2 = 40$.

We then calculate our p-value using the command below. Our p-value, .1309, is greater than .05. Hence, we do not reject the null hypothesis that the population slope coefficient is .2.

[10]:
```
2 * t.cdf(-1.548, 40)
```

[10]:
```
0.1295
```

By the same logic, the t statistic to test a null hypothesis about the population intercept is:

$$t = \frac{b_0 - \beta_{0,0}}{s_{b_0}}$$

In the above expression, $\beta_{0,0}$ denotes the null hypothesis for the intercept and $s_{b_0}$ is the estimated standard error of the estimated intercept. The number of degrees of freedom is the same as for the slope coefficient.

Suppose we wish to test the null hypothesis that a population coefficient is zero. This test is automated for us when we estimate a regression, and the associated p-values are provided in the fourth column of the regression output. We can easily verify this as follows. Suppose we wish to test the null hypothesis that the population slope coefficient is zero for our regression above. Proceeding as above, we calculate the t-statistic:

$$t = \frac{.1356 - 0}{.024} = 5.65$$

The regression reports the t value 5.551. The difference between our calculation and the result in the regression output is roundoff error. The following command then yields a p-value barely above zero. Look back to the regression, and you will find the p-value for the slope coefficient is $4.57e-06$. Our p-value below is the same as the value in the regression output, .0000. The p-value is far below our significance level of .05, so we reject the null hypothesis that the population slope coefficient is zero.

[11]:
```
2 * t.cdf(-5.65, 40)
```

[11]:
```
0.0000
```

**Hypothesis Tests Reported As Part of Regression Output** Why does the software automatically test the null hypothesis that the population coefficients are zero? When we estimate a regression, we generally do so with the expectation that the right-hand-side variable is important

in predicting the left-hand-side variable. If we reject the hypothesis that the population slope coefficient is zero, then we have evidence that the population coefficient is not equal to zero. This is good news in most applications.[4]

Above, we tested the null hypothesis that the population slope coefficient is .2. If you look back at the regression output, you will see that Python reports a 95 percent confidence interval for each coefficient. The 95 percent confidence interval for the slope coefficient is [.086,.185]. We see that .2 lies in this interval. Hence, we immediately see that we reject the null hypothesis of .2 at significance level $\alpha = .05$.

Not all software packages report confidence intervals for coefficients. Hence, it is important that you know how to test a null hypothesis using the approach we laid out above for calculating the t-statistic for the test and the associated p-value.

**Key criteria for Evaluating a simple regression model:**

- **Sign**: Is the algebraic sign of the slope coefficient as expected? There may be times when you don't have an *a priori* expectation regarding sign, but often you will.
- **Significance**: Is the estimate statistically significant?
- **Magnitude**: Is the magnitude of the coefficient plausible? In the context of your application, is it a small, moderate, or large coefficient? If the estimated coefficient is implausibly large or small, you will want to rethink your understanding of the issue you are studying.

**Applying The Criteria to Our Application** We see above that the slope coefficient, the coefficient of Year, is .1356.

**Sign:** Is the algebraic sign of the slope as expected?

- Yes. We expect to find that area burned has been increasing over time.

**Significance:** Is the estimated coefficient of year statistically significant at the 5% level?[5]

- Yes. As we noted above, the p-value is barely above zero, hence well below .05. Thus, we strongly reject the hypothesis that the population slope coefficient is zero.

**Magnitude:** Interpret the coefficient. Is it a small, moderate, or large effect?

- The coefficient tells us that for each additional year, acreage burned is predicted to increase by 135600 acres. This strikes me as a large annual increase in acres of forest burned, but, sadly, not at all implausible.

**Further Thoughts About Interpretation** Let's now study other features of the regression. The intercept is negative. Should we worry about this?

- No. The intercept is the result we obtain when we extrapolate all the way back to year zero, an extrapolation back in time of 1,984 years. It would be foolish to take seriously an extrapolation so extraordinarily far outside of the range of our data. To get more insight into this issue, look back at the plot of the data and the regression line. We see that the red line

[4] Of course in this application, we would have preferred to find that area burned is not increasing over time.] When we reject the null hypothesis that the coefficient equals zero, we say that the coefficient is **statistically significant**.

[5] It is relatively standard practice to use significance level $\alpha = .05$ in judging whether a result is statistically significant.

is positive throughout the range of our data, i.e., predicted values are positive over the range of our data.

What fraction of the variation in Acres burned is explained by the Year variable?

- The $R^2$ statistic tells us that 35% of the variation is explained by the Year variable.

Do the above results imply that this is a causal relationship?

- No. The passage of time need not cause more forests to burn.
- This is a descriptive model. It tells us that acres of forest burned has increased over the time period from 1984 through 2025, but it does not tell us why. Later in the course, we will study a model of forest area burned in California that takes account of several factors that affect area burned.

As we noted above, our software calculates confidence intervals for coefficients. Given that some software packages do not report confidence intervals, it is important for you to know how to do it. You can calculate a confidence interval "by hand" using the information in the regression output. For example, the confidence interval for the slope coefficient is given by:

$$[b_1 - s_{b_1} * t_{\alpha/2,\nu}, b_1 + s_{b_1} * t_{\alpha/2,\nu}]$$

Let's calculate the confidence interval for the coefficient of Year in the regression above. We obtain $t_{\alpha/2,\nu}$ using the command below and obtain 2.03.

[12]:
```
t.ppf(0.975, 40)
```

[12]:
```
2.0211
```

The remaining information we need is in the regression output above. Our confidence interval is: $[.1356 - .024 * 2.021, .1356 + .024 * 2.021]$, or $[.087, .184]$.
Look back at the regression output and you will see that, up to roundoff error, this confidence interval for the slope coefficient is as reported in the regression output.

**Prediction** Let's calculate the predicted value of Acres burned for 1985 using our regression. We use the coefficients in our regression and plug in 1985 as our value of Year. Our predicted value is then (-266.4386+.1356*1985)=2.727 million acres. This is the point on the regression line directly above 1985 in the graph presented earlier in this chapter.

Here is some additional terminology. When we calculate a predicted value for an observation used to estimate the model, it is typically called a **fitted value**. Thus, in the previous paragraph, we calculated the fitted value of Acres=2.727 when Year=1985.

In many applications, our objective in estimating a model is to do prediction outside the sample used for estimation. Our sample contains data through 2025. Suppose we wish to investigate the model's prediction of acreage burned in 2026. Let's do the prediction and calculate a 95% prediction interval. Note the format of the predict_interval() command. We enter the name of the regression to be used for the prediction. We then enter a dataframe with the name of the right-hand-side variable from the regression and the value of the variable to be used for the prediction. We then specify the confidence level for the prediction. By default, the cofidence level is set to .95. I use a confidence level of .95 below.

[13]:
```
# Reminder:prediction_intervals expects a dataframe
Xnew = pandas.DataFrame({"Year": [2026]})
RegAcres.prediction_intervals(Xnew)
```

[13]:

| | Predicted | Lower | Upper |
|---|---|---|---|
| 0 | 8.359995 | 3.450401 | 13.26959 |

The model predicts that 8.36 million acres were burned in 2022, with a 95% prediction interval [3.45,13.27] million acres. The prediction interval tells us that there is a 95% probability that area burned in year 2026 will be between 3.45 and 13.27 million acres. Not surprisingly, the prediction interval is quite wide. As we found in our regression analysis, there is a significant upward trend in acreage burned, but there is also much variability from year to year.

You may be wondering why the interval is called a prediction interval rather than a confidence interval. When we do a prediction, there are two sources of uncertainty. 1) We do not know the population coefficients and must estimate them. 2) We do not know the error term $\varepsilon$. The prediction interval takes account of both of these sources of uncertainty.

As noted above, the model developed above is not a causal model. Models that are not causal can sometimes be useful for prediction. However, we need to keep in mind that such a model may cease to give valid predictions if the underlying causal factors change.

### 1.5.2 Application 2: Temperature as a Function of Year

We have established that there was a large and statistically significant increase in forest acreage burned over the period between 1985 and 2020. Let's next investigate whether there was a significant change in global average temperature over this time period. The regression is shown below.

As you study the regression, please think about the following questions.

- Is the algebraic sign of coefficient of year as you would expect?
- Is the coefficient statistically significant at the 5% level?
- What is the interpretation of the coefficient of Tempf? Do you think this is a large effect.
- What fraction of the variation in temperature between 1985 and 2020 is explained by Year?
- Is this a causal relationship?

As with our previous regression, the model relating time and temperature is a descriptive rather than a causal relationship. It establishes that there has been significant global warming over this time period. The estimated increase in temperature per year over this time period is .035 $^oF$ per year.

[14]:
```
RegTemp = tprstats.model("ts", "Tempf~Year", FTD84)
RegTemp.summary()
```

[14]:

| | | | |
|---|---|---|---|
| **Dep. Variable:** | Tempf | **R-squared:** | 0.885 |
| **Model:** | OLS | **Adj. R-squared:** | 0.882 |
| **No. Observations:** | 42 | **F-statistic:** | 182.2 |
| **Covariance Type:** | HAC | **Prob (F-statistic):** | 1.76e-16 |

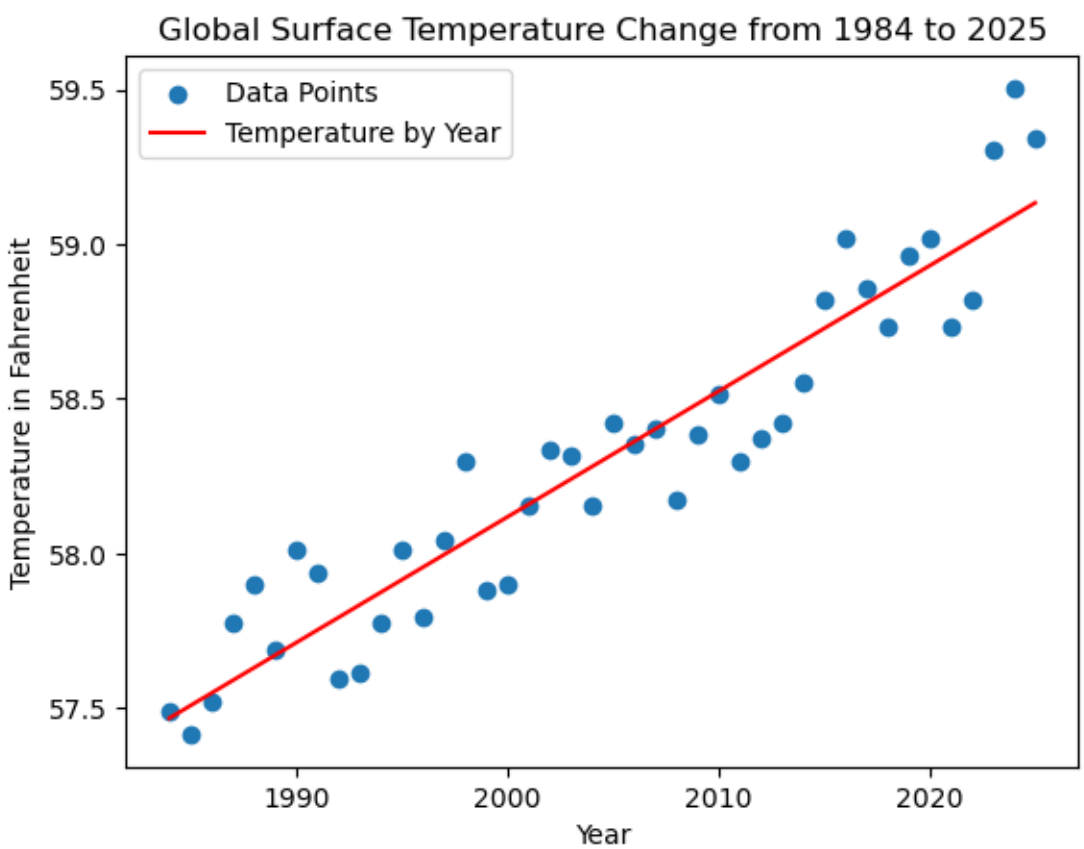
| | coef | std err | t | P> \|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **Intercept** | -23.1236 | 6.023 | -3.839 | 0.000 | -35.297 | -10.950 |
| **Year** | 0.0406 | 0.003 | 13.498 | 0.000 | 0.035 | 0.047 |

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 2 lags and without small sample correction
[2] The condition number is large, 3.32e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

Let's plot the relationship between temperature and year and include the regression line.

[15]:
```
# Predicted values
y_pred = RegTemp.predict()

# plt.figure(figsize=(10, 6))
plt.scatter(FTD84["Year"], RegTemp.y, label="Data Points")
plt.plot(FTD84["Year"], y_pred, color="red", label="Temperature by Year")
plt.xlabel("Year")
plt.ylabel("Temperature in Fahrenheit")
plt.title("Global Surface Temperature Change from 1984 to 2025")
plt.legend()
plt.show()
```



### 1.5.3 Application 3: Acres Burned as a Function of Temperature

Estimate a regression of acres burned as a function of temperature. The result is shown below. As we study this regression, thinking about **sign**, **significance**, and **magnitude**, we see:

- The algebraic sign of the coefficient of Tempf is positive, as we would expect.
- The coefficient is statistically significant at the 5% level.
- The regression predicts that a .1 degree increase in temperature increases forest burned by 2,744,300 acres.

Is this a causal relationship? We would certainly expect that higher temperatures would increase forest area burned. However, factors other than temperature may affect forest area burned such as precipitation and human activities. Hence, we would not want to claim that this is a causal model. Later in the course we will study a model that takes account of several factors that affect area burned in California. We will find that temperature is one of those factors.

[16]:
```
RegAcresTemp = tprstats.model("ts", "US_Acres_Burned~Tempf", FTD84)
RegAcresTemp.summary()
```

[16]:

| | | | |
|---|---|---|---|
| **Dep. Variable:** | US_Acres_Burned | **R-squared:** | 0.264 |
| **Model:** | OLS | **Adj. R-squared:** | 0.246 |
| **No. Observations:** | 42 | **F-statistic:** | 14.36 |
| **Covariance Type:** | HAC | **Prob (F-statistic):** | 0.000498 |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **Intercept** | -154.5486 | 42.109 | -3.670 | 0.001 | -239.655 | -69.442 |
| **Tempf** | 2.7443 | 0.724 | 3.790 | 0.000 | 1.281 | 4.208 |

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 2 lags and without small sample correction
[2] The condition number is large, 6.5e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

Let's plot the relationship between acres burned and temperature and include the regression line. I've added explanatory text to each command to help you understand all of the commands. To help interpret the graph, I've chosen a value of temperature from our data set, 58.73 $^oF$. This was the temperature in year 2021. I placed a dashed vertical line at that temperature. The black triangle where the dashed line intersects the regression line is the predicted area burned when temperature is 58.73. That predicted area is 6.22 million acres. The actual area burned that year, 7.126 million acres, is the blue circle immediately above the black triangle on the dashed line. The difference between the actual and predicted is .503 million acres. This is called the residual and is represented by the solid orange line.The same temperature, 58.73 $^oF$ occurred in year 2018. The area burned was much higher in 2028. The is the higher blue circle on the vertical dashed line.

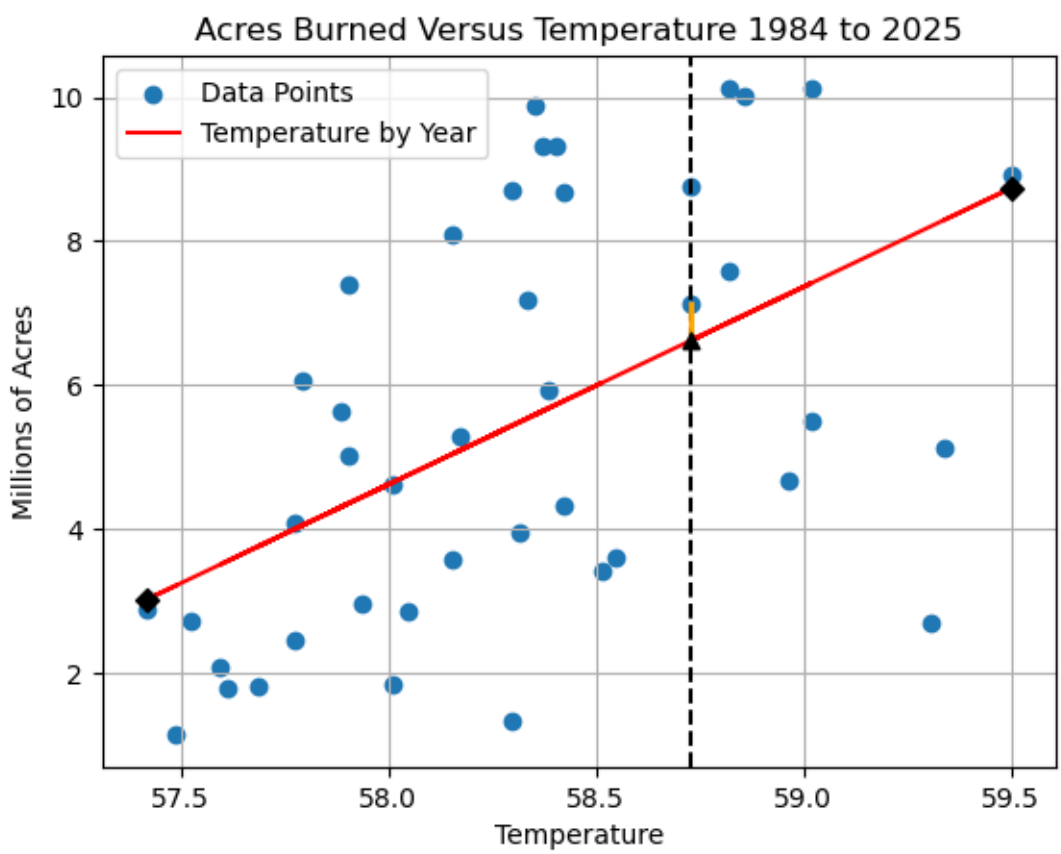
[17]:
```
y_pred = RegAcresTemp.predict()
```

```
# Plot
plt.axvline(x=58.73, color="black", linestyle="--") # vertical line

plt.plot([58.73, 58.73], [6.622, 7.126], color="orange", linewidth=2,␣
↪linestyle="-") # residual line

# plt.figure(figsize=(10, 6))
plt.scatter(FTD84["Tempf"], RegAcresTemp.y, label="Data Points")
plt.plot(FTD84["Tempf"], y_pred, color="red", label="Temperature by Year") #␣
↪regression line
plt.xlabel("Temperature")
plt.ylabel("Millions of Acres")
plt.title("Acres Burned Versus Temperature 1984 to 2025")
plt.legend()
plt.scatter(59.504, 8.746, color="black", zorder=5, marker="D")
plt.scatter(57.416, 3.016, color="black", zorder=5, marker="D")
plt.scatter(58.73, 6.622, color="black", zorder=5, marker="^")


plt.grid(True)
plt.show()
```



Earlier in these notes, we defined the fitted values, $\hat{y}_i$ from a regression:

$$\hat{y}_i = b_0 + b_1 x_i$$

We also defined the estimated residuals:

$$e_i = y_i - \hat{y}_i$$

The following commands show how to calculate the residuals and the fitted values. Note where the name of the regression, RegAcresTemp, is used in these commands.

[18]:
```
ResidAcres = RegAcresTemp.resid
PredAcres = RegAcresTemp.predict()
```

Let's put Year, Tempf, Acres, PredAcres, and ResidAcres in a spreadsheet for viewing. As you can see below, I named the spreadsheet OurData. We found from the summary statistics earlier in this chapter that the lowest temperature in the data was in 1985, 57.416$^o$. We see from the spreadsheet that the predicted acreage burned in 1985 was 3.016 million acres. This predicted value is shown by the black diamond at the lower end of the regression line in the above graph. We also found that the highest temperature in our data was 59.50$^o$ degrees, which is for year 2024. From our spreadsheet, you will see that the predicted acreage burned was 8.746 million acres in 2024. This predicted value is shown as a black diamond at the upper end of the regression line in the graph above. Thus, the model predicts that the rise in temperature from 1985 through 2024 is associated with a (8.746-3.016)=5.73 million increase in acres burned.

From the spreadsheet, we see that the predicted value for the observation with temperature 59.504 is 8.746. This is shown by a black triangle on the regression line in the above plot. The residual for this observation is .178. It is the vertical difference between the dot above the triangle and the triangle. In the spreadsheet below, you will find that these values are for year 2024.

[19]:
```
# Python
OurData = FTD84[["Year", "Tempf", "US_Acres_Burned"]].copy()
OurData["ResidAcres"] = ResidAcres
OurData["PredAcres"] = PredAcres

OurData
```

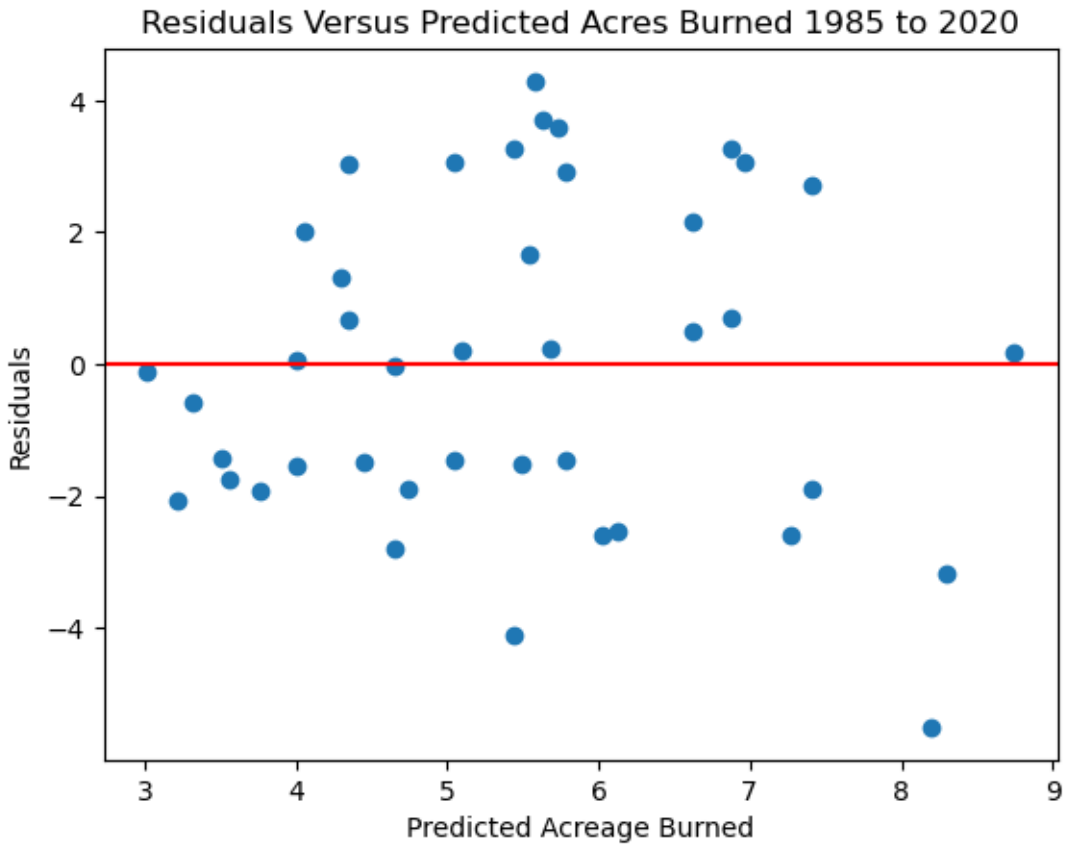
[19]:

| | Year | Tempf | US_Acres_Burned | ResidAcres | PredAcres |
|---|---|---|---|---|---|
| 104 | 1984 | 57.488 | 1.148409 | -2.065498 | 3.213907 |
| 105 | 1985 | 57.416 | 2.896147 | -0.120172 | 3.016319 |
| 106 | 1986 | 57.524 | 2.719162 | -0.593538 | 3.312700 |
| 107 | 1987 | 57.776 | 2.447296 | -1.556960 | 4.004256 |
| 108 | 1988 | 57.902 | 5.009290 | 0.659256 | 4.350034 |
| 109 | 1989 | 57.686 | 1.827310 | -1.929962 | 3.757272 |
| 110 | 1990 | 58.010 | 4.621621 | -0.024794 | 4.646415 |
| 111 | 1991 | 57.938 | 2.953578 | -1.495249 | 4.448827 |

```
112  1992  57.596        2.069929   -1.440359   3.510288
113  1993  57.614        1.797574   -1.762111   3.559685
114  1994  57.776        4.073579    0.069323   4.004256
115  1995  58.010        1.840546   -2.805869   4.646415
116  1996  57.794        6.065998    2.012345   4.053653
117  1997  58.046        2.856959   -1.888249   4.745208
118  1998  58.298        1.329704   -4.107060   5.436764
119  1999  57.884        5.626093    1.325456   4.300637
120  2000  57.902        7.393493    3.043459   4.350034
121  2001  58.154        3.570911   -1.470678   5.041589
122  2002  58.334        7.184712    1.649154   5.535558
123  2003  58.316        3.960842   -1.525319   5.486161
124  2004  58.154        8.097880    3.056291   5.041589
125  2005  58.424        8.689389    2.906847   5.782542
126  2006  58.352        9.873745    4.288790   5.584955
127  2007  58.406        9.328045    3.594900   5.733145
128  2008  58.172        5.292468    0.201482   5.090986
129  2009  58.388        5.921786    0.238038   5.683748
130  2010  58.514        3.422724   -2.606802   6.029526
131  2011  58.298        8.711367    3.274603   5.436764
132  2012  58.370        9.326238    3.691887   5.634351
133  2013  58.424        4.319546   -1.462996   5.782542
134  2014  58.550        3.595613   -2.532707   6.128320
135  2015  58.820       10.125149    3.255877   6.869272
136  2016  59.018        5.509995   -1.902642   7.412637
137  2017  58.856       10.026086    3.058020   6.968066
138  2018  58.730        8.767492    2.145204   6.622288
139  2019  58.964        4.664364   -2.600083   7.264447
140  2020  59.018       10.122336    2.709699   7.412637
141  2021  58.730        7.125643    0.503355   6.622288
142  2022  58.820        7.577183    0.707911   6.869272
143  2023  59.306        2.693910   -5.509076   8.202986
144  2024  59.504        8.924884    0.178533   8.746351
145  2025  59.342        5.131474   -3.170306   8.301780
```

Plot the residuals versus fitted values and add a horizontal line. This type of plot can be useful in detecting whether the relationship is linear. In this case, there is no obvious nonlinearity.

[20]:
```
# plt.figure(figsize=(10, 6))
plt.scatter(OurData["PredAcres"], OurData["ResidAcres"])
plt.axhline(y=0, color="red", label="y = 0")
plt.xlabel("Predicted Acreage Burned")
plt.ylabel("Residuals")
plt.title("Residuals Versus Predicted Acres Burned 1985 to 2020")
plt.show()
```




**Prediction** We now do prediction using this regression. It is possible with a single command to do predictions for multiple values of the right-hand-side variable. Here is an example. It seems likely that surface temperatures will continue to rise. Let's predict forest area burned for three possible future temperature values, 59, 59.25, 59.5. Note the format of the predict command.

[21]:
```
RegAcresTemp.predict(pandas.DataFrame({"Tempf": [59, 59.25, 59.5]}))
```

[21]:
```
0    7.363240
1    8.049307
2    8.735374
dtype: float64
```

We see that an increase in temperature from 59 $^oF$ to 59.5 $^oF$ is predicted to increase area burned by (8.05-7.36)=.65 million acres.

**What Changes if We Express Temperature in Celsius Rather than Fahrenheit?** We estimated the regression equation above using temperature measured in Fahrenheit. Suppose we had instead estimated the model using temperature in degrees Celsius. Our data file has a column named Tempc. This is temperature in degrees Celsius. For practice, estimate the regression of Acres versus Tempc. Then, note that the t-statistic and p-value for the coefficient of Tempc are

the same as for the coefficient of Tempf in our regression above. Note also that the t-statistic and p-value for the intercept are different. The following provides the explanation.

We know that $F = \frac{9}{5}C + 32$. Our estimated regression with temperature in Fahrenheit is: $\hat{y} = -154.549 + 2.744 * F$. Substitute for $F$ and obtain $\hat{y} = -154.549 + 2.744 * (\frac{9}{5}C + 32)$. Simplify and obtain: $\hat{y} = -66.732 + 4.940 * C$. Using the data in FTD8, estimate the regression of acres with Tempc as the explanatory variable. Up to round off error, you should get the coefficient estimates we just derived.

My reason for the exercise in the preceding paragraph is to illustrate that the magnitude of a coefficient depends on the unit of measurement. One $^oC$ is 9/5 larger than one $^oF$. As a result the slope coefficient when using $^oC$ is 9/5 larger than when using $^oF$. Note also that the t-statistic and p-value for the slope coefficient are the same whether we use Fahrenheit or Celsius. This is because the t-statistic is scale free, and the p-value is calculated using the t-statistic. Here is the intuition: Our null hypothesis is that the population slope coefficient is zero. If the slope coefficient is zero when temperature is measured in $^oF$, then the slope must be zero if temperature were measured in $^oC$.

In addition, note that the t-statistic and p-value change for the intercept. Why? When we change from Fahrenheit to Celsius, we add (9/5)*32 to the intercept. If this effect of changing the intercept is a bit elusive to you, don't worry about it. The key point is for you to understand why the slope coefficient changed as it did and why the significance of the slope coefficient did not change.

### 1.5.4 Application 4: GMAT and GPA in Courses at Tepper

In this application, we investigate how well total GMAT predicts graduating GPA in required courses at Tepper. Let RGPA be a shorthand for GPA in required courses. The data set for this application is for graduates several years ago. To preserve confidentiality of the data, I converted both variables to percentiles. In particular, I created GMAT_pctile as the percentile ranking of the student's entering GMAT total score. Similarly, for each student, I created RGPA_pctile as the percentile score of each graduating student. The data are in an Excel file named GPA_Data.xlsx.

**Import GPA_Data and Estimate Regression**

[22]:
```
GPA_Data = pandas.read_excel("data/GPA_Data.xlsx")
```

Viewing the regression output, we see that the coefficient of GMAT is positive as expected. From the p-value, we see that the coefficient is statistically significant. The magnitude tells us that a one percentage point increase in GMAT predicts a .378 percentage point increase in graduating GPA in required courses. From the $R^2$ statistic, we see that this regression can account for 20% of the variation in graduating GPA in required courses. You may find it surprising that the $R^2$ is so low. Of course, this is a quite simple model. Later in the course, we will explore whether the quantitative and verbal scores have differential predictive value, and we will also investigate other variables such as work experience prior to attending Tepper.

[23]:
```
RegRGPA = tprstats.model("cs", "RGPA_pctile~GMAT_pctile", GPA_Data)
RegRGPA.summary()
```

[23]:

| | | | |
|---|---|---|---|
| **Dep. Variable:** | RGPA_pctile | **R-squared:** | 0.204 |
| **Model:** | OLS | **Adj. R-squared:** | 0.203 |
| **No. Observations:** | 624 | **F-statistic:** | 169.3 |
| **Covariance Type:** | HC1 | **Prob (F-statistic):** | 2.08e-34 |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **Intercept** | 31.0329 | 1.787 | 17.363 | 0.000 | 27.523 | 34.543 |
| **GMAT_pctile** | 0.3781 | 0.029 | 13.012 | 0.000 | 0.321 | 0.435 |

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

### 1.6 Conditions for Obtaining Unbiased Coefficient Estimates

When we estimate a regression model, we wish to obtain unbiased estimates of the population coefficients. A key requirement for the estimated slope coefficient to be unbiased is that $\varepsilon$ in equation (1) be uncorrelated with the right-hand-side variable $x$. The intercept will also be unbiased if the preceding condition is satisfied and the expected value of the error terms is zero. That is $E(\varepsilon_i) = 0$. This brings us back to the **Gauss Markov Theorem**. As we go along in the course, we will study applications that will clarify the meaning of these requirements for obtaining unbiased estimates.

### 1.7 Summary

In this chapter, we have seen how to estimate and evaluate simple regression models using four applications. These applications provide a review of material you learned in Probability and Statistics. Please review the learning objectives at the outset of these notes to be sure that you have a good grasp of the topics covered in this chapter.

Also, I hope you find the topics we study in these notes to be interesting. Throughout the course, I try to provide applications that are informative about statistical methods while also being of of practical interest.

## 1.8 Appendix: Derivation of Formulas To Estimate Coefficients in Simple Regression

This appendix provides the derivation of the formulas that we use to estimate the coefficients in the simple regression model. **I strongly emphasize that this is optional.** I present it in case you wish to see the mathematical derivation of the formulas.

The coefficients can be derived using matrix algebra. The matrix derivation is simpler than the derivation below, but the matrix derivation may not be as transparent as the derivation below. While the algebraic details below are a bit tedious, these details may be helpful to you in understanding the logic and implementation of the approach.

### 1.8.1 Definition of The Sum of Squared Errors

Equation (1) defines the estimated error term in the simple regression model given values for the coefficients $b_0$ and $b_1$. Subscript $i$ denotes an observation in a sample of data. Our goal is to derive formulas for $b_0$ and $b_1$ that minimize the sum of squared errors.

1. $e_i = (y_i - b_0 - b_1 x_i)$

Let $n$ denote the number of observations in the sample. The sum of squared errors is then obtained by squaring the expression in equation (1) and summing the result across the $n$ observations.

2. $\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$

### 1.8.2 Deriving the Formulas to Estimate the Coefficients

To minimize the sum of squared errors, we take the derivatives of the expression in equation (2) with respect to $b_0$ and $b_1$. We then set the derivatives to zero to find the values that minimize the sum of squared errors.

**Derivative with Respect to the Intercept** The derivative with respect to $b_0$ is as follows. The equality of the second and third terms in equation (3) follows from the property that the derivative of a sum equals the sum of the derivatives.

3. $\frac{\partial \sum_{i=1}^{n} e_i^2}{\partial b_0} = \frac{\partial \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2}{\partial b_0} = \frac{\sum_{i=1}^{n} \partial (y_i - b_0 - b_1 x_i)^2}{\partial b_0} = \sum_{i=1}^{n} [-2(y_i - b_0 - b_1 x_i)] = -2 \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i) = 0$

We now rewrite the last expression from the above equation. Canceling $-2$, we obtain:

4. $\sum_{i=1}^{n} (y_i - b_0 - b_1 x_i) = \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} b_0 - b_1 \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i - n b_0 - b_1 \sum_{i=1}^{n} x_i = 0$

Using the last expression from equation (4) and dividing by $n$, we obtain:

5. $\frac{\sum_{i=1}^{n} y_i}{n} - b_0 - b_1 \frac{\sum_{i=1}^{n} x_i}{n} = \overline{y} - b_0 - b_1 \overline{x} = 0$

Solving (5) for $b_0$ yields the following.

6. $b_0 = \overline{y} - b_1 \overline{x}$

To use the formula in equation (6) to calculate $b_0$, we need to derive the equation for $b_1$. We next undertake that derivation.

**Derivative with Respect to the Slope Coefficient** Substitute (6) into (2) and collect terms to obtain:

7. $\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n}(y_i - \overline{y} + b_1\overline{x} - b_1 x_i)^2 = \sum_{i=1}^{n}((y_i - \overline{y}) - b_1(x_i - \overline{x}))^2$

The derivative with respect to $b_1$ is as follows. The equality of the second and third terms in equation (8) follows, as above, from the fact that the derivative of a sum equals the sum of the derivatives.

8. $\frac{\partial \sum_{i=1}^{n} e_i^2}{\partial b_1} = \frac{\partial \sum_{i=1}^{n}((y_i-\overline{y})-b_1(x_i-\overline{x}))^2}{\partial b_1} = \frac{\sum_{i=1}^{n} \partial((y_i-\overline{y})-b_1(x_i-\overline{x}))^2}{\partial b_1} = -2\sum_{i=1}^{n}[((y_i-\overline{y})-b_1(x_i-\overline{x}))(x_i-\overline{x})] = 0$

Cancel -2 in (8) and rearrange to obtain:

9. $\sum_{i=1}^{n}[(y_i - \overline{y})(x_i - \overline{x}) - b_1(x_i - \overline{x})(x_i - \overline{x})] = \sum_{i=1}^{n}(y_i - \overline{y})(x_i - \overline{x}) + b_1\sum_{i=1}^{n}(x_i - \overline{x})(x_i - \overline{x}) = 0$

Solve (9) for $b_1$:

10. $b_1 = \frac{\sum_{i=1}^{n}(y_i-\overline{y})(x_i-\overline{x})}{\sum_{i=1}^{n}(x_i-\overline{x})(x_i-\overline{x})} = \frac{\sum_{i=1}^{n}(y_i-\overline{y})(x_i-\overline{x})}{\sum_{i=1}^{n}(x_i-\overline{x})^2}$

We can calculate the estimated slope coefficient $b_1$ using Equation (10). They, we substitute the estimate of $b_1$ into Equation (6) and calculate $b_0$. These are our least squares estimates of the intercept and slope in the simple regression model.

The above concludes our derivation of the equations for the intercept and slope coefficients. The following provides a useful interpretation of the equation for the slope coefficient.

Divide the numerator and denominator of (10) by (n-1).

11. $b_1 = \frac{\frac{\sum_{i=1}^{n}(y_i-\overline{y})(x_i-\overline{x})}{(n-1)}}{\frac{\sum_{i=1}^{n}(x_i-\overline{x})^2}{(n-1)}}$

The numerator in (11) is the sample covariance of $x$ and $y$. We denote this $s_{xy}$. The denominator is the sample variance of $x$, which we denote $s_x^2$.

Hence, we can rewrite the slope coefficient in (11) as the ratio of the sample covariance of $x$ and $y$ to the sample variance of $x$.

12. $b_1 = \frac{s_{xy}}{s_x^2}$

### 1.8.3 Practice Example

For practice, let's estimate the intercept and slope for a regression using the following five data points.

x 1, 2, 3, 4, 5

y 2, 2, 4, 4, 8

By eyeballing the data, you can easily see that $\overline{x}$=5 and $\overline{y}$=4. Similarly, by eyeballing the data, you can easily obtain the deviations from the means of $x$ and $y$. For example, from above, $x_1$=1. The mean of $x$ is 5. Hence, $(x_1 - \overline{x})$=-4. The deviations of the $x_i$ from $\overline{x}$ for the five data points are then as follows:

$(x_i - \overline{x})$ : -4, -2, 0, 2, 4

The deviations of the $y_i$ from $\overline{y}$ for the five data points are as follows:
$(y_i - \overline{y})$ : : -2, -2, 0, 0, 4

Using the above, the sample covariance between x and y is:

$s_{xy} = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{n-1}$=28/4

The sample variance of x is:

$s_x^2 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}$=40/4

From equation (12), $b_1$ is the ratio of the above. Hence, $b_1$=(28/4)/(40/4)=.7

From equation (6), $b_0 = \overline{y} - b_1\overline{x}$=4-.7*5=.5

Thus, we find the estimated intercept is .5 and the estimated slope is .7.

### 1.8.4 Some Useful Terminology

The equations we use to calculate the intercept and slope are called **estimators**. The numerical values we obtain when we plug our data into the the formulas are called **estimates**.